Programming Assignment

Name: Amna Malik ID: R11827604

Problem-1

(Your Codes)

```java
package parts;
import java.util.Scanner;

/**
 * Converts a number from 1 to 90 into Roman numerals.
 */
public class parta {

    /*
     * Turns a number into Roman numerals
     */
    public static String toRoman(int number) {

        // Stores final Roman numeral
        String roman = "";

        // Roman numeral values
        int[] values = {90, 50, 40, 10, 9, 5, 4, 1};

        // Roman numeral symbols
        String[] numerals = {"XC", "L", "XL", "X", "IX", "V", "IV", "I"};

        // Builds the Roman numeral string
        for (int i = 0; i < values.length; i++) {
            while (number >= values[i]) {
                roman += numerals[i];
                number -= values[i];
            }
        }

        return roman;
    }

    /*
     * Runs the program and handles user input/output
     */
    public static void main(String[] args) {

        // Used for input
        Scanner scanner = new Scanner(System.in);

        // Stores user number
        int number;

        // Keep asking until valid
        while (true) {
            System.out.print("Enter a number 1 to 90: ");
```

```
            number = scanner.nextInt();

            if (number >= 1 && number <= 90) {
                break;
            }

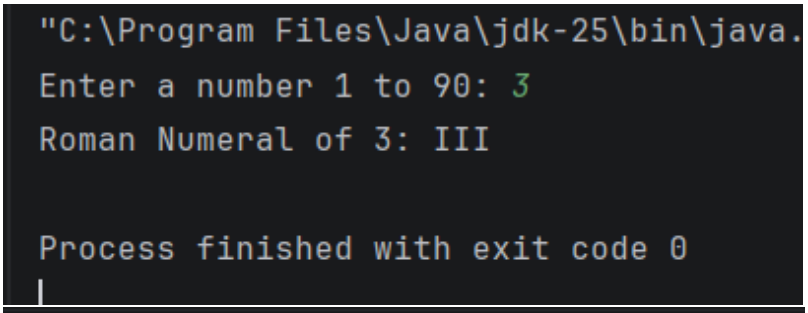
            System.out.println("Sorry, that choice is invalid. Please try
again.");
        }

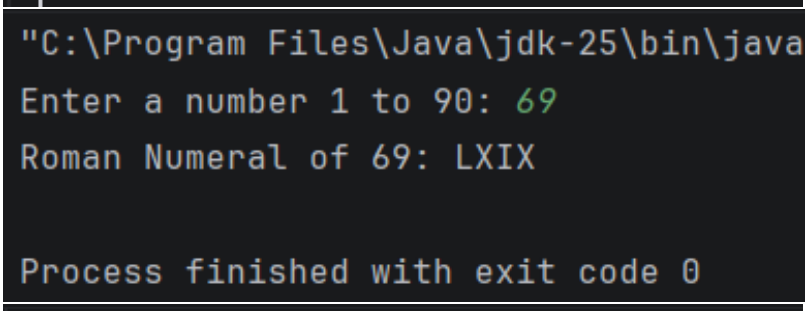
        // Stores Roman numeral result
        String romanNumeral = toRoman(number);

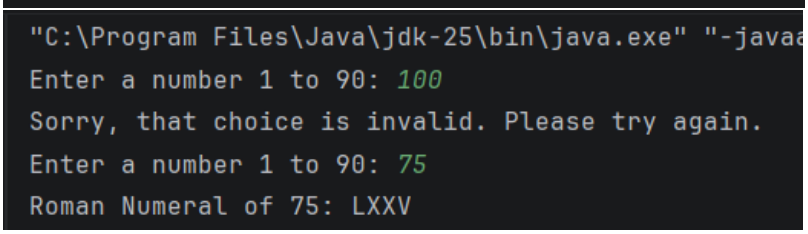
        // Prints result
        System.out.println("Roman Numeral of " + number + ": " +
romanNumeral);

        scanner.close();
    }
}
```

(Output- screenshots)







Problem-2

(Your Codes)

```java
package parts;

import java.util.Scanner;

/**
 * Takes 3 names and prints them in alphabetical order.
 */
public class partb {

    /*
     * Sorts 3 strings in ascending order
     */
    public static String[] sort3(String a, String b, String c) {

        // Stores the names in an array
        String[] names = {a, b, c};

        // Simple sort (3 items)
        if (names[0].compareToIgnoreCase(names[1]) > 0) {
            String temp = names[0];
            names[0] = names[1];
            names[1] = temp;
        }

        if (names[1].compareToIgnoreCase(names[2]) > 0) {
            String temp = names[1];
            names[1] = names[2];
            names[2] = temp;
        }

        if (names[0].compareToIgnoreCase(names[1]) > 0) {
            String temp = names[0];
            names[0] = names[1];
            names[1] = temp;
        }

        return names;
    }

    /*
     * Runs the program and handles user input/output
     */
    public static void main(String[] args) {

        // Used for input
        Scanner scanner = new Scanner(System.in);

        // Stores first name
        System.out.print("Enter the first name: ");
        String name1 = scanner.nextLine();
```

```
        // Stores second name
        System.out.print("Enter the second name: ");
        String name2 = scanner.nextLine();

        // Stores third name
        System.out.print("Enter the third name: ");
        String name3 = scanner.nextLine();

        // Stores sorted names
        String package parts;

import java.util.Scanner;

/**
 * Gets 3 names and prints them in alphabetical order.
 */
public class partb {

    /*
     * Sorts three names A to Z
     */
    public static String[] sortNames(String a, String b, String c) {

        // Holds the three names
        String[] names = {a, b, c};

        // Swap first two if needed
        if (names[0].compareToIgnoreCase(names[1]) > 0) {
            String temp = names[0];
            names[0] = names[1];
            names[1] = temp;
        }

        // Swap last two if needed
        if (names[1].compareToIgnoreCase(names[2]) > 0) {
            String temp = names[1];
            names[1] = names[2];
            names[2] = temp;
        }

        // Check first two again
        if (names[0].compareToIgnoreCase(names[1]) > 0) {
            String temp = names[0];
            names[0] = names[1];
            names[1] = temp;
        }

        return names;
    }

    /*
     * Runs the program and prints sorted names
     */
    public static void main(String[] args) {

        // Used for input
```

```
        Scanner scanner = new Scanner(System.in);

        // First name
        System.out.print("Type the first name: ");
        String first = scanner.nextLine();

        // Second name
        System.out.print("Type the second name: ");
        String second = scanner.nextLine();

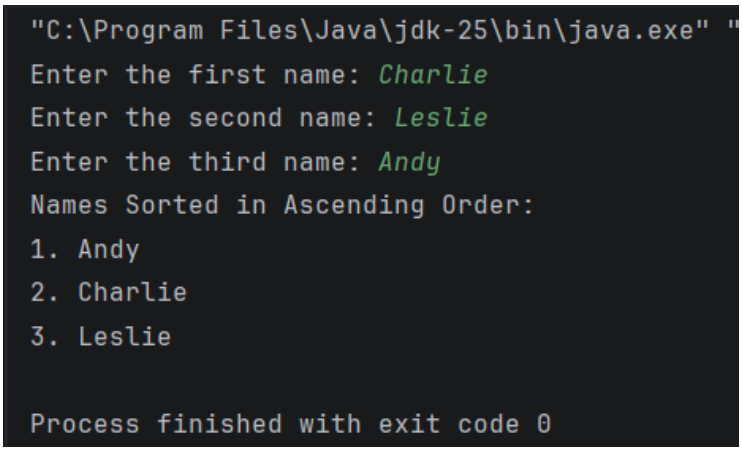
        // Third name
        System.out.print("Type the third name: ");
        String third = scanner.nextLine();

        // Sorted result
        String[] sorted = sortNames(first, second, third);

        // Print output
        System.out.println("Here are the names in order:");
        System.out.println("1. " + sorted[0]);
        System.out.println("2. " + sorted[1]);
        System.out.println("3. " + sorted[2]);

        scanner.close();
    }
}
```

(Output- screenshots)



```
"C:\Program Files\Java\jdk-25\bin\java.exe"
Enter the first name: Tooba
Enter the second name: Amna
Enter the third name: Tahir
Names Sorted in Ascending Order:
1. Amna
2. Tahir
3. Tooba

Process finished with exit code 0
```

Problem-3

(Your Codes)

```java
package parts;

import java.util.Scanner;

/**
 * Helps walk through fixing Wi-Fi problems.
 */
public class partc {

    /*
     * Checks if input means yes
     */
    public static boolean saidYes(String input) {

        // Cleaned input
        String text = input.trim().toLowerCase();

        return text.equals("yes") || text.equals("y");
    }

    /*
     * Runs the troubleshooting steps
     */
    public static void main(String[] args) {

        // Used for input
        Scanner scanner = new Scanner(System.in);

        // User response
        String reply;

        // Step 1
        System.out.println("Reboot the computer and try to connect.");
        System.out.print("Did that fix the problem? ");
        reply = scanner.nextLine();

        if (saidYes(reply)) {
            scanner.close();
            return;
        }

        // Step 2
        System.out.println("Restart the router and try to connect.");
        System.out.print("Did that fix the problem? ");
        reply = scanner.nextLine();

        if (saidYes(reply)) {
            scanner.close();
            return;
        }

        // Step 3
```

```
        System.out.println("Check that the modem and router cables are
plugged in.");
        System.out.print("Did that fix the problem? ");
        reply = scanner.nextLine();

        if (saidYes(reply)) {
            scanner.close();
            return;
        }

        // Step 4
        System.out.println("Move the router somewhere else.");
        System.out.print("Did that fix the problem? ");
        reply = scanner.nextLine();

        if (saidYes(reply)) {
            scanner.close();
            return;
        }

        // Final step
        System.out.println("You may need a new router.");

        scanner.close();
    }
}
```

(Output-screenshots)

